

NABC PRODUCT PITCH

SentryBand

A Wearable Real-Time Heat Stress & Toxic Gas
Exposure Alert System for Marine Engine Room Personnel

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GED106: Purposive Communication — Product Pitch



The Engine Room Is Already the Hardest Watch



Extreme heat, by design

Engine rooms are consistently ranked among the hottest, most physically taxing spaces on a ship — heat radiation, poor ventilation, non-stop exertion.



Fatigue dulls the warning signs

Research shows heat and fatigue measurably reduce a crew member's ability to notice a developing problem before it turns dangerous.



Training alone isn't enough

Safety certification builds hazard knowledge — but it can't compensate for a slow, cumulative rise in heat strain a person can't feel in the moment.

A Monitoring Gap That Costs Lives

~350

seafarers & shore workers lost to enclosed-space asphyxiation since 1996

70

of those deaths occurred in just the last 3 years — the trend hasn't slowed

~95%

of fatal ferry incidents occur on smaller, domestic-route vessels

Portable gas detectors exist — but only for planned inspections, not worn through a routine watch. Smaller, domestic vessels rarely carry the fixed, class-approved monitoring larger ships already have.

APPROACH

One Wristband, Two Hazards Sensed



Heat-strain sensing

Skin temperature & heart rate, worn against the wrist.



Toxic-gas sensing

CO, hydrogen sulfide & combustible vapors in the surrounding air.



Threshold logic

ESP32 microcontroller checks readings against NIOSH & gas limits.



Wireless relay

Bluetooth Low Energy link to the engine control room dashboard.

SentryBand: Component Architecture Overview

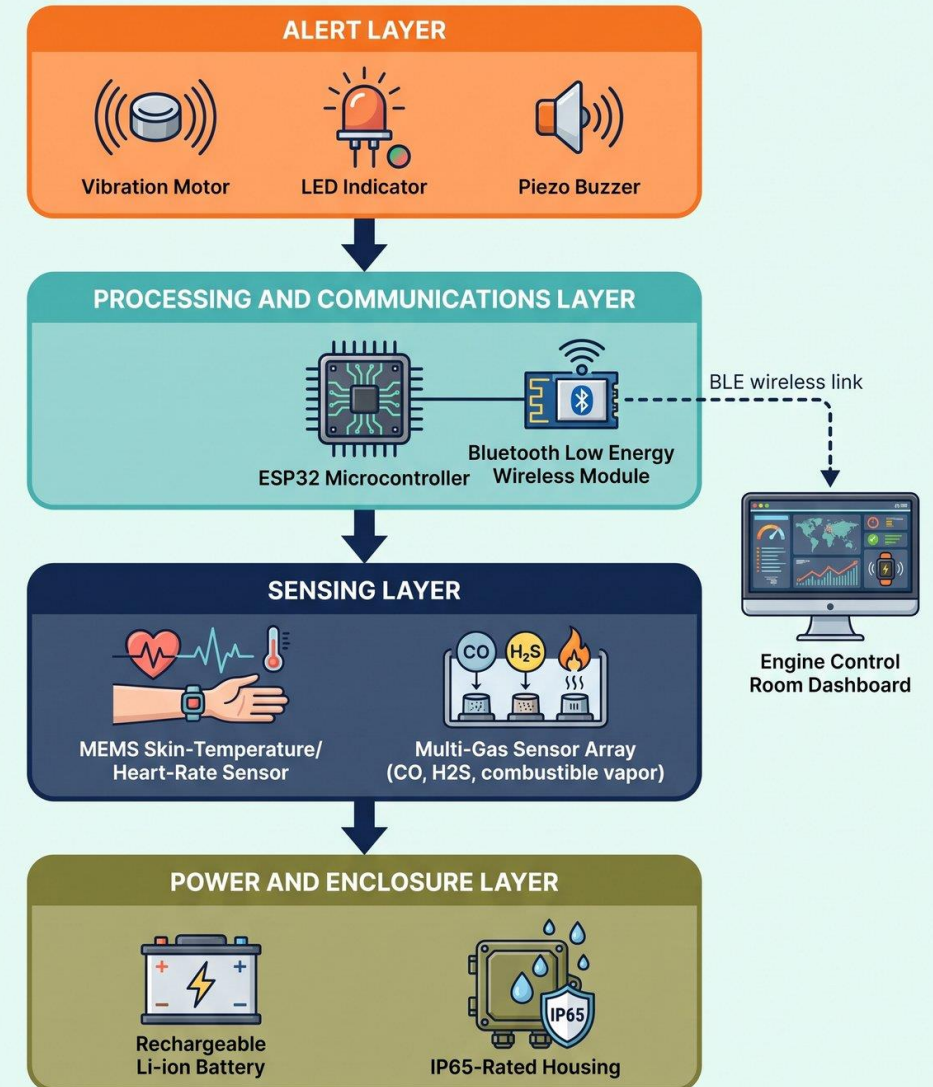


Diagram Credit: Higgsfield AI

APPROACH

A Continuous Sense → Compare → Alert Loop

Every watch runs the same five-step cycle:

1. **Activation & Calibration** — baseline readings recorded.
2. **Continuous Sensing** — sensors sample at fixed intervals.
3. **Threshold Comparison** — checked against NIOSH & gas limits.
4. **Tiered Alerting** — vibration, then full alarm + dashboard.
5. **Logging & Review** — trends available after the watch.

SentryBand: Process Flow Summary

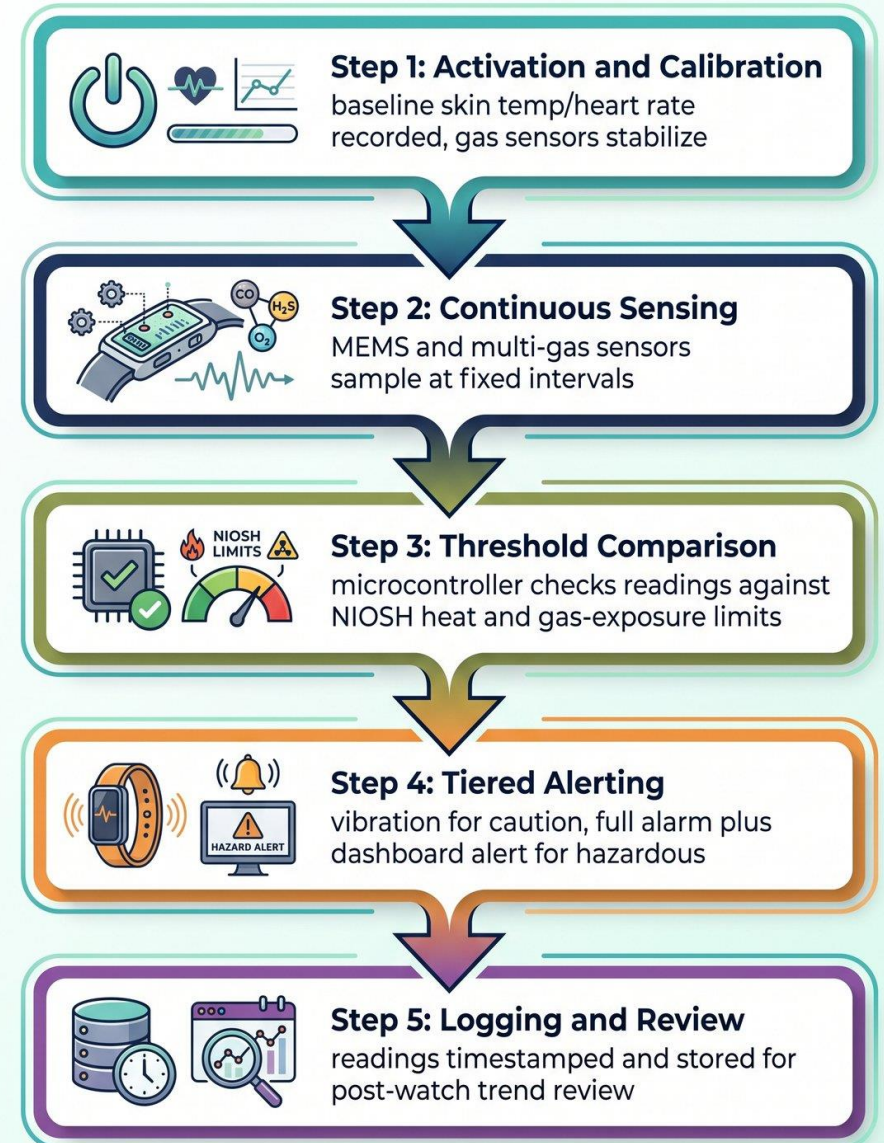


Diagram Credit: Higgsfield AI

Built for the Engine Room Itself

Step 3: Seal and Attach the Wearable Strap

Seal the assembled electronics into the water- and oil-resistant housing, then attach the adjustable wristband strap.



IP65-rated housing

Heat- and oil-resistant enclosure, sealed against dust and water jets.



Full-watch battery

Rechargeable Li-ion battery sized to run an entire multi-hour shift.



Retrofits into routine

Straps on like any watch — no changes to existing shipboard systems.

Fig. Seal & Attach — Assembly, Technical Report

Warned Before It Happens — Not After



Early, dual warning

Both the wearer and the officer of the watch are alerted before a heat collapse or gas-exposure incident happens.



One device, two risks

The only approach that watches heat strain and gas risk together, continuously, in a single low-cost unit.



Logged for review

Every reading is timestamped, so the chief engineer can spot recurring hot spots or gas-prone areas.

Why SentryBand Over What Exists Today

Fixed Monitoring Systems

- ✓ Effective, class-approved
- ✗ Costly to install & certify
- ✗ Only on larger vessels

Portable Gas Detectors

- ✓ Reliable readings
- ✗ Used only for planned entry
- ✗ Not worn all shift

Wearable Health Trackers

- ✓ Proven low-cost sensing
- ✗ No gas sensing at all
- ✗ Not engine-room rated

SentryBand

- ✓ Continuous, dual sensing
- ✓ Engine-room rated (IP65)
- ✓ Priced for small/medium vessels

SentryBand

A personal guardian for every engine room watch.

Thank you.

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